B.Sc. (Hons.) in Environmental Science

- B.Sc. (Hons.) in Environmental Science is an undergraduate degree program that focuses on the study of the environment and its interactions with living and non-living components.
- The program typically lasts for three years and covers a wide range of topics, including ecology, conservation biology, environmental chemistry, and environmental policy.
- Students enrolled in this program will learn about the scientific principles underlying environmental issues and will develop the skills necessary to analyze and solve complex environmental problems.
- The program includes both theoretical and practical components, with students participating in laboratory work, fieldwork, and research projects.
- Graduates of this program may pursue careers in environmental consulting, conservation, research, or policy-making.
- Many universities offer B.Sc. (Hons.) in Environmental Science programs, and admission requirements typically include a strong background in science and mathematics.

Unit 1 - Environment

Definition

The environment is the sum total of all the physical, chemical, and biological factors and conditions that surround and influence an organism or group of organisms.

Types of Environment

There are two main types of environment: natural and built. The natural environment includes all living and non-living things that occur naturally on Earth, such as air, water, soil, plants, and animals. The built environment includes all human-made elements, such as buildings, roads, and other infrastructure.

Components of Environment

The environment is made up of several components, including: - The atmosphere: the layer of gases surrounding the Earth - The hydrosphere: all the water on Earth, including oceans, lakes, rivers, and groundwater - The lithosphere: the solid outer layer of the Earth, including rocks and soil - The biosphere: all living organisms on Earth, including plants, animals, and microorganisms

Segments of Environment

The environment can be divided into several segments, including: - The terrestrial environment: the land and all the living and non-living things on it - The aquatic environment: all the water on Earth and the living and non-living things in it - The atmospheric environment: the air and all the living and non-living things in it

Scope and Importance

The environment is essential for the survival and well-being of all living organisms. It provides the resources we need to live, such as food, water, and shelter. It also plays a crucial role in regulating the Earth's climate and maintaining the balance of ecosystems.

Need for Public Awareness

It is important for the public to be aware of the environment and the impact of human activities on it. This can help people make informed decisions about how to live sustainably and reduce their impact on the environment.

Ecosystem

Definition

An ecosystem is a community of living and non-living things interacting with each other in a specific environment.

Types of Ecosystem

There are many different types of ecosystems, including: - Terrestrial ecosystems: ecosystems on land, such as forests, grasslands, and deserts - Aquatic ecosystems: ecosystems in water, such as oceans, lakes, and rivers - Human-made ecosystems: ecosystems created by humans, such as urban areas and agricultural land

Structure of Ecosystem

An ecosystem is made up of several components, including: - Producers: organisms that produce their own food, such as plants - Consumers: organisms that eat other organisms, such as animals - Decomposers: organisms that break down dead organic matter, such as fungi and bacteria - Abiotic factors: non-living elements that influence the ecosystem, such as sunlight, water, and soil

Food Chain

A food chain is a sequence of organisms in an ecosystem where each organism feeds on the one below it. For example, in a grassland ecosystem, the food chain might be: grass -> rabbit -> fox.

Food Web

A food web is a more complex representation of the feeding relationships in an ecosystem. It shows how multiple food chains are interconnected.

Ecological Pyramid

An ecological pyramid is a graphical representation of the trophic levels in an ecosystem. It shows the relative amount of energy or biomass at each trophic level.

Balance Ecosystem

A balanced ecosystem is one where all the components are in harmony and there is no overpopulation or depletion of resources.

Effects of Human Activities on Environment

Human activities, such as food production, shelter, housing, agriculture, industry, mining, transportation, economic and social security, can have a significant impact on the environment. These activities can lead to pollution, deforestation, loss of biodiversity, and climate change.

Environmental Impact Assessment

Environmental Impact Assessment (EIA) is a process used to evaluate the potential environmental impacts of a proposed project or development. It helps decision-makers determine whether a project should proceed and what measures should be taken to minimize its impact on the environment.

Sustainable Development

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It aims to balance economic, social, and environmental concerns to ensure a sustainable future for all.

Unit 2 - Natural Resources: Introduction, Classification

Natural resources are materials and substances that occur naturally within the environment and can be used for economic gain. They are classified into two main categories: renewable and non-renewable resources.

Renewable resources are those that can be replenished naturally in a relatively short period of time, such as sunlight, wind, and water. Non-renewable resources, on the other hand, are those that exist in a fixed amount and cannot be replenished within a human lifespan, such as fossil fuels and minerals.

Water Resources

Water is a vital resource for all living organisms and is essential for many human activities, including agriculture, industry, and domestic use. The availability of water resources varies greatly depending on geographic location, climate, and human activities.

Sources of water include surface water, such as rivers and lakes, and groundwater, which is found beneath the Earth's surface. The quality of water is also an important consideration, as it can be affected by pollution and contamination.

Water-borne diseases are illnesses caused by microorganisms that are transmitted through contaminated water. These can include cholera, typhoid, and dysentery. Water-induced diseases, on the other hand, are illnesses caused by the lack of clean water or sanitation, such as dehydration and skin infections.

Fluoride and arsenic are two common contaminants found in drinking water. High levels of fluoride can cause dental and skeletal fluorosis, while high levels of arsenic can cause skin lesions and increase the risk of cancer.

Mineral Resources

Minerals are naturally occurring substances that are extracted from the Earth for their economic value. They include metals, such as gold and copper, and non-metallic minerals, such as sand and gravel.

Material Cycles

Material cycles refer to the natural processes by which elements and compounds are cycled through the environment. These include the carbon, nitrogen, and sulfur cycles.

The carbon cycle involves the movement of carbon between the atmosphere, oceans, and land. Carbon dioxide is taken up by plants during photosynthesis and is released back into the atmosphere through respiration and decomposition.

The nitrogen cycle involves the movement of nitrogen between the atmosphere, soil, and living organisms. Nitrogen is converted into various forms, such as ammonia and nitrate, through processes such as nitrogen fixation and nitrification.

The sulfur cycle involves the movement of sulfur between the atmosphere, soil, and living organisms. Sulfur is released into the atmosphere through volcanic eruptions and the burning of fossil fuels, and is taken up by plants and returned to the soil through decomposition.

Energy Resources

Energy resources are sources of energy that can be used to generate electricity, heat, and motion. They can be classified into two main categories: conventional and non-conventional sources of energy.

Conventional sources of energy are those that have been widely used for many years, such as coal, oil, and natural gas. Non-conventional sources of energy, on the other hand, are those that are relatively new and not yet widely used, such as solar, wind, and geothermal energy.

Forest Resources

Forests are a valuable natural resource that provides timber, fuel, and other products. They also play an important role in regulating the Earth's climate and water cycle.

The depletion of forests can have a significant impact on the environment and society. It can lead to soil erosion, loss of biodiversity, and changes in local weather patterns. It can also affect the livelihoods of people who depend on forests for their survival.

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution is the introduction of harmful substances or products into the environment. It is a major problem in America and as well as the world. Pollution not only damages the environment, but it also damages the health of the people. There are several types of pollution, and while they may come from different sources and have different consequences, understanding the basics about pollution can help environmentally conscious individuals minimize their contribution to these dangers.

Water Pollution

Water pollution is the contamination of water bodies, usually as a result of human activities. Water bodies include for example lakes, rivers, oceans, aquifers and groundwater. Water pollution results when contaminants are introduced into the natural environment. For example, releasing inadequately treated wastewater into natural water bodies can lead to degradation of aquatic ecosystems. In turn, this can lead to public health problems for people living downstream. They may use the same polluted river water for drinking or bathing or irrigation.

Air Pollution

Air pollution is the release of pollutants into the air that are detrimental to human health and the planet as a whole. The Clean Air Act authorizes the U.S. Environmental Protection Agency (EPA) to protect public health by regulating the emissions of these harmful air pollutants. The NRDC has been a leading authority on this law since it was established in 1970.

Soil Pollution

Soil pollution is caused by the presence of man-made chemicals or other alteration in the natural soil environment. This type of contamination typically arises from the rupture of underground storage tanks, application of pesticides, percolation of contaminated surface water to subsurface strata, oil and fuel dumping, leaching of wastes from landfills or direct discharge of industrial wastes to the soil. The most common chemicals involved are petroleum hydrocarbons, solvents, pesticides, lead and other heavy metals.

Noise Pollution

Noise pollution is the disturbing or excessive noise that may harm the activity or balance of human or animal life. The source of most outdoor noise worldwide is mainly caused by machines and transportation systems, motor vehicles, aircraft, and trains.

Solid Waste Management

Solid waste management is the collection, transport, processing, recycling or disposal, and monitoring of waste materials. The term usually relates to materials produced by human activity, and the process is generally undertaken to reduce their effect on health, the environment or aesthetics. Waste management is a distinct practice from resource recovery which focuses on delaying the rate of consumption of natural resources. All waste materials, whether they are solid, liquid, gaseous or radioactive fall within the remit of waste management.

Unit 4 - Current Environmental Issues of Importance

1. **Global Warming**: This is an increase in the ambient temperature of Earth which will negatively affect life on Earth. Human activities contribute to global warming by increasing the greenhouse effect.

2. **Green House Effects**: The greenhouse effect happens when certain gases—known as greenhouse gases —collect in Earth's atmosphere. The greenhouse gases responsible for the greenhouse effect are: Water Vapour, Carbon Dioxide, Methane, and Ozone. Scientists have determined that carbon dioxide's warming effect helps stabilize Earth's atmosphere.

3. **Climate Change**: Climate change is the long-term alteration in Earth's climate and weather patterns, primarily as a result of increased levels of carbon dioxide and other pollutants caused by human activities.

4. **Acid Rain**: Acid rain is a type of precipitation that contains high levels of sulfuric or nitric acids. The effects of acid rain, combined with other environmental stressors, leave trees and plants less healthy, more vulnerable to cold temperatures, insects, and disease.

5. **Ozone Layer Formation and Depletion**: The ozone layer is a region of Earth's stratosphere that contains a high concentration of ozone (O3) molecules. It protects the Earth from the harmful ultraviolet (UV) radiation from the Sun. However, human activities such as the release of chlorofluorocarbons (CFCs) have caused the depletion of the ozone layer.
6. **Population Growth**: Population growth is the increase in the number of individuals in a population. It can have both positive and negative impacts on the environment. On one hand, it can lead to increased consumption of resources and production of waste. On the other hand, it can also drive innovation and technological advancements that can help mitigate environmental issues.
7. **Automobile pollution**: Automobiles emit pollutants such as carbon monoxide, nitrogen oxides, and particulate matter that can have negative impacts on air quality and human health.
8. **Burning of paddy straw**: The burning of paddy straw, also known as rice straw, is a common practice in some agricultural regions. However, it can release pollutants into the air and contribute to air pollution.

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act of 1986 is an act of the Parliament of India.
- It was enacted to provide for the protection and improvement of the environment.
- The act empowers the central government to establish authorities charged with the mandate of preventing environmental pollution in all its forms.
- The act also provides for the creation of a National Green Tribunal to adjudicate on environmental disputes.

Initiatives by Non Governmental Organizations (NGO's)

- Non-governmental organizations (NGOs) play a crucial role in environmental protection.
- They work to raise awareness about environmental issues and promote sustainable development.
- NGOs also work to influence government policies and actions related to the environment.
- Some well-known NGOs working in the field of environmental protection include Greenpeace, World Wildlife Fund, and Friends of the Earth.

Human Population and the Environment

Population growth

- The rapid growth of the human population is one of the biggest challenges facing the environment.
- Population growth leads to increased demand for resources, which can result in overexploitation and degradation of the environment.
- To address this issue, it is important to promote sustainable development and responsible resource use.

Environmental Education

- Environmental education is the process of teaching individuals about the environment and how to protect it.
- It aims to increase awareness and understanding of environmental issues and promote positive action to address them.
- Environmental education can take place in formal and informal settings, and can target people of all ages.

Women Education

- Educating women is crucial for environmental protection.
- Women often play a central role in managing natural resources and promoting sustainable development.
- Educating women can empower them to make informed decisions about resource use and to participate in environmental decision-making.